

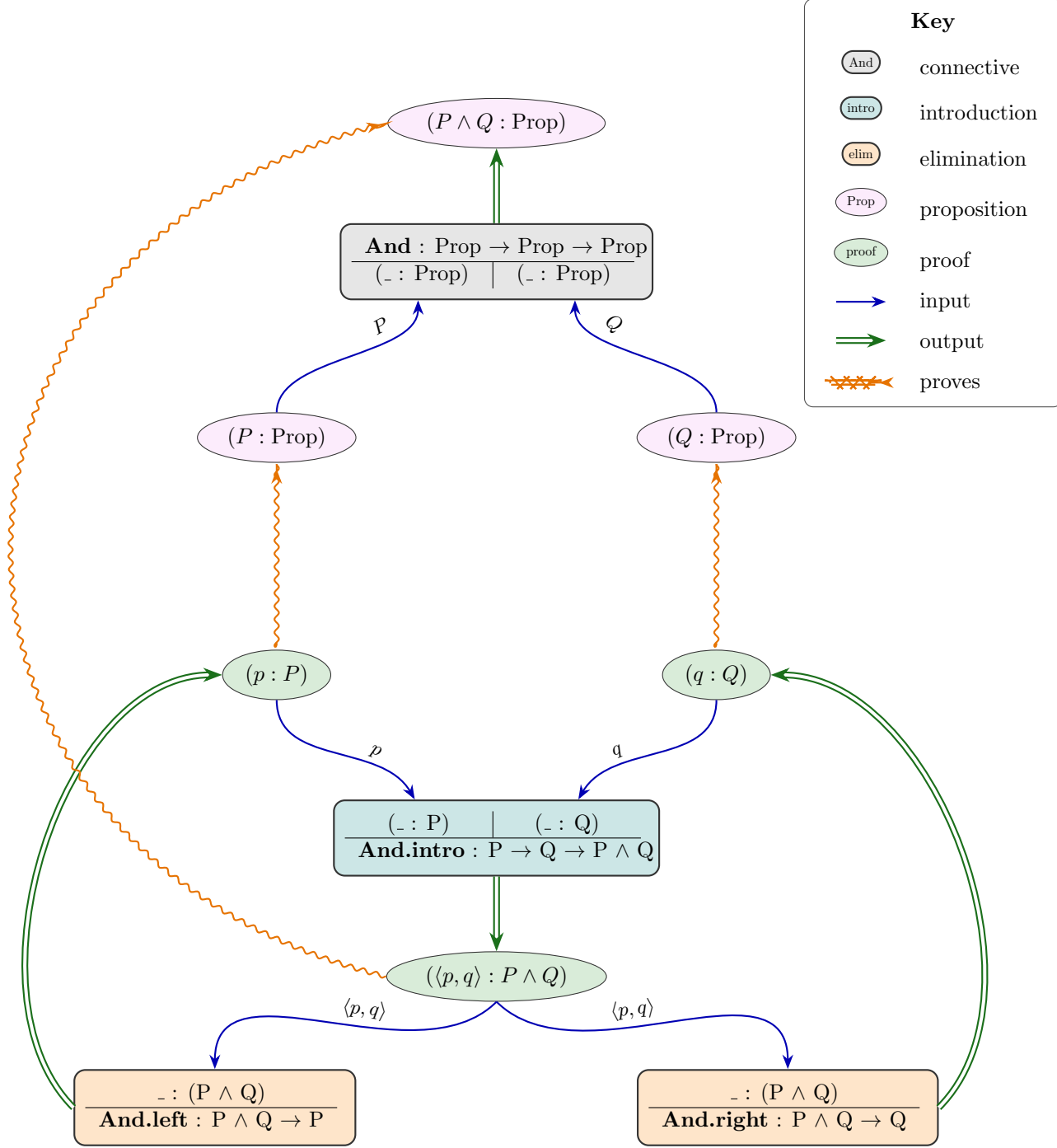


Figure 1: The algebra of logical conjunction. The And function takes two propositions, call them P and Q , and yields the proposition that is their logical conjunction, $P \wedge Q$. The *And.intro* function takes two proofs p of P and of q of Q and returns a proof, $\langle p, q \rangle$ of their conjunction. The *And.left* function takes a proof, $\langle p, q \rangle$ of $P \wedge Q$, and returns the proof, $(p : P)$, while *And.right* returns $q : Q$. The proof. All of these proof handling functions implicitly take the types P and Q themselves as implicit arguments. They are typically not provided explicitly when these functions are applied to their arguments, but are inferred automatically from the types of these arguments; and we've elided them from this diagram.